

PHD SEMINAR · DAY 1 · CASE STUDY 1

Financial Fraud Detection

Scale, Taxonomy, Data & Sources

The concrete problem that grounds everything from **Foundations of Machine Learning and AI in Finance**: how big fraud is, how it's classified, where the public data lives, and how to cite it correctly.

Why fraud, specifically?

1

Extreme class imbalance

Fraud is often $< 1\%$ of transactions — as extreme as this challenge gets.

2

Concept drift

Fraud schemes evolve constantly, faster than most other financial prediction problems.

3

Weak, delayed labels

"Fraud" is confirmed only after investigation — sometimes months after the transaction.

4

Regulatory & interpretability stakes

Flagging a customer wrongly, or missing fraud, both carry real accountability.

The approach in this case study: ground the theory in real numbers and real (public) data before we touch a single classifier. That comes next, in the Classification Under Imbalance session.

A taxonomy of financial fraud

"Fraud detection" spans several distinct problems with different data, labels, and stakeholders



Occupational fraud

Employees defrauding their own organization: asset misappropriation, corruption, billing schemes.



Payment / card fraud

Unauthorized card-present or card-not-present transactions; the classic "transaction-level" ML problem.



Financial statement fraud

Deliberately misstated corporate financials — rarest but most costly per ACFE data.



Insurance fraud

Inflated or fabricated claims; strong asymmetric-cost framing (investigate vs. honor).

This case study focuses primarily on payment/card fraud — the most data-rich, most-studied of the four.

How big is the problem? Occupational fraud

ACFE, Occupational Fraud 2024: A Report to the Nations — 1,921 real cases, 138 countries

5%

of annual revenue lost to fraud,
on average, per organization

\$1.7M

median total loss per case studied

89%

of cases were asset
misappropriation — most
common, least costly

5%

of cases were financial statement
fraud — rarest, most costly
(median \$766K)

43% of frauds were caught by tips — 3x any other detection method — and median losses rose 24% from 2022 to 2024, ending three prior cycles of decline.

Source: Association of Certified Fraud Examiners, Occupational Fraud 2024: A Report to the Nations (acfe.com)

Two more angles: cards & corporate crime

Payment card fraud

\$33.41B

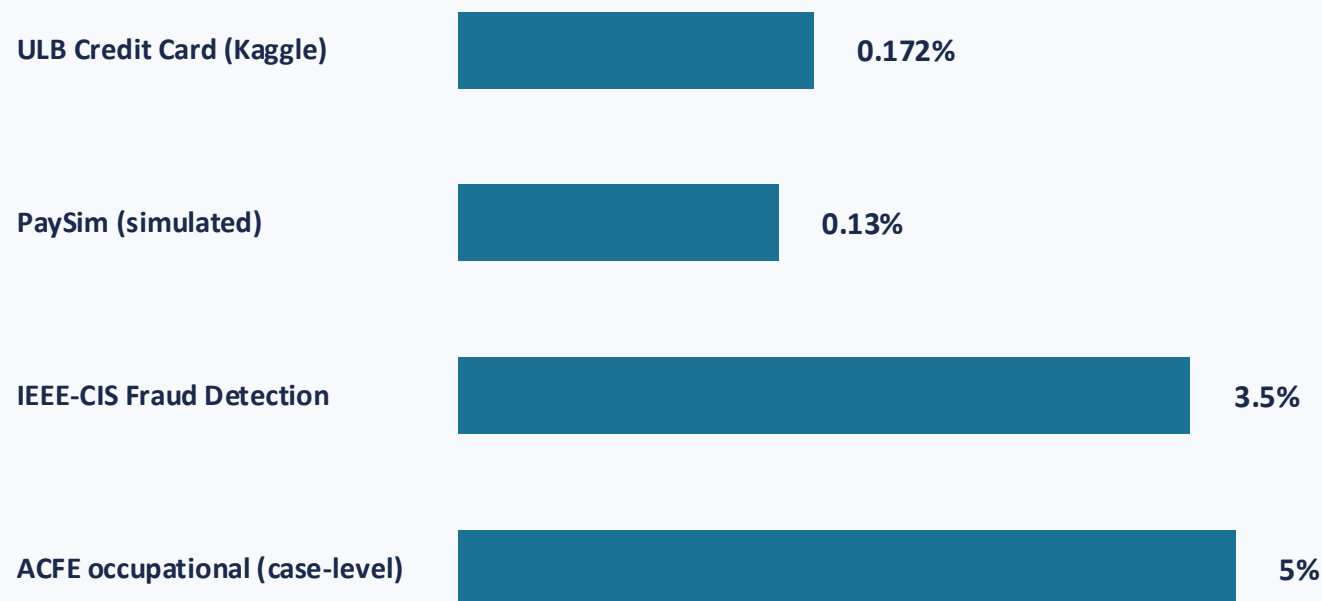
worldwide card fraud losses in 2024, tied to \$51.92T in global card volume

Projected to reach **\$41.06B by 2030**. The US accounts for ~26% of global card volume but ~42% of global fraud losses — a useful reminder that fraud rates vary sharply by market.

Source: The Nilson Report, Issue 1298 (Jan 2026)

The public dataset landscape

Fraud rate on a log scale — notice how far left these sit compared to a "balanced" 50%



bar length $\propto \log_{10}(\text{fraud rate})$ — a linear chart would make the first two bars invisible

Dataset notes

ULB (Kaggle): 284,807 EU card transactions, 492 fraud, 30 PCA-anonymized features.

PaySim: 6.3M simulated mobile-money transactions — watch for label leakage in the balance columns.

IEEE-CIS: 590,540 real e-commerce transactions, hundreds of features.

ACFE occupational (case-level): 1,921 real investigated cases, 138 countries — case-level, not transaction-level; no feature vectors, only aggregated case attributes (scheme type, loss amount, detection method).

Also worth knowing: Bank Account Fraud (BAF) and Healthcare Provider Fraud datasets, for non-card domains.

??? Reading these sources critically

Not all "fraud statistics" are produced — or should be cited — the same way

Source type	Example	What to check before citing
Industry survey report	<i>ACFE, Nilson, PwC</i>	Sampling method, respondent self-selection, whether figures are extrapolated or directly observed
Peer-reviewed literature review	<i>Reurink (2018), Journal of Economic Surveys</i>	Scope of the review, publication dates, how "fraud" is defined across the studies it synthesizes
Peer-reviewed empirical paper	<i>Fu et al. (2025), Procedia CS</i>	Dataset provenance, sample size, whether results replicate outside that one dataset
Dataset documentation	<i>Kaggle/IEEE-CIS dataset cards</i>	Collection window, known leakage columns, anonymization method, license terms

This case study's exercises

1

Notebook: dataset landscape + bias-variance demo

- Compare fraud rates and scale across the public datasets on a single chart.
- Run the guided under/good/overfit demo from Module 1 and connect it back to real fraud data.
- Reflection cells: which dataset would you pick for a first project, and why?

2

Briefing memo: source literacy

- In pairs, take one report or dataset from today's slides.
- Extract 3 key numbers and one methodological caveat from the source-literacy table.
- Write a one-page memo (template provided) and deliver a 5-minute briefing to the group.

Key takeaways

- ✓ Fraud spans distinct sub-problems (occupational, card, statement, insurance) with different data and stakeholders.
- ✓ Every credible figure here traces to a named, dated source — and every source type demands a different kind of scrutiny.
- ✓ Public fraud datasets sit at the extreme end of class imbalance: 0.1–5% positive class is the norm, not the exception.
- ✓ This scale and structure is exactly what classification techniques are built to handle.

Looking ahead — September 22

Tomorrow: Fraud classification. We put this scale to work — build a classifier on imbalanced data, watch the accuracy trap happen live, and fix it with resampling, cost-sensitive learning, and threshold tuning — then move into ensembles, anomaly detection, and time-series methods.